Kapasa Makasa University

Department of Agriculture and Aquatic Sciences

Agribusiness Management (SA: 500)

Lecture 8: Financial Decision Making in Agribusiness

by

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Introduction to Decision-Making

- Decision-making is the **process** of choosing between different alternatives for the purpose of achieving desired goals.

Elements of Decision-Making

Element	Meaning
Process	Active, systematic procedure
Choosing	Selecting among feasible alternatives
Goals	Clear, measurable objectives

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Objectives

By the end of Lecture 8, you should be able to:

- Describe the decision-making process
- Apply operating decision-making tools (Revenue & Cost Analysis, Input-Cost Analysis & Cost-Volume Profit Analysis)
- Apply investment decision-making tools (Payback Period, Simple Rate of Return & Net Present Value (NPV)
- Discuss various Sources of Finance for Agribusiness
- Agribusiness Risk Mgt & Insurance

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The Decision-Making Process (Steps)

The decision-making process is simply a logical procedure for identifying a problem, analysing it, and arriving at a solution.

1. Problem Identification
2. Summary of Facts
3. Identifying Alternatives
4. Analysis
5. Action
6. Evaluation

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The Decision-Making Process (Steps)

1. Problem Identification

This is often the toughest part of decision-making. It is easy to confuse symptoms with the real problem. The problem may seem to be **low profits**, when low profits are simply the result of **an inefficient, high-cost distribution** system. Once the problem is clearly defined, it can usually be more easily addressed.

3. Identifying Alternatives

This step identifies and lists **feasible alternative solutions,** exploring various possibilities. Only feasible solutions should be considered.

2. Summary of Facts

This step brings to the surface and **highlights information pertinent to the problem and its solution**. It may be critical to note overall company goals, the impact of the problem on the business, environmental factors limiting possible solutions, or technical facts that affect the outcome.

4. Analysis

This step may require rigorous **examination**, **weighing the costs** and **benefits** of each alternative. Analysis considers both short and long-term company goals. Although analysis should be objective, the final selection process should include some subjective evaluation of alternatives.

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The Decision-Making Process (Steps)

5. Action

One of the most critical steps in the decision-making process involves implementing the chosen alternative. Often this requires careful planning prior to the execution, but it is a critical step. Management responsibility goes much further than just deciding; it requires execution and results.

6. Evaluation

The final step in this process occurs some period of time after action has been taken. Management must evaluate whether or not the firm is better off by the action taken. If the situation has improved, no further action is necessary. However, if the action taken has not caused the desired results, management must go through this process again and seek new action alternatives.

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Types of Financial Decision-Making in Agribusiness

Agribusiness Financial Decisions include two main types:

1. **Operating Decisions:** decisions that have a shorter (usually one year or less) time frame.
2. **Investment Decisions:** decisions with impacts that will last much longer than a single year.

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1. Operating Decisions

Operating Decision-Making tools

The following are the operating decision-making tools:

(i) Cost-Volume Profit or breakeven analysis.

(ii) Revenue and Cost Analysis

(iii) Input–output analysis

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(i) Cost-Volume Profit Analysis

Cost-Volume Profit Analysis, or **breakeven analysis**, is a tool for examining the relationship between costs, the volume and profits of a business generated by the firm.

It shows:

- The level of volume necessary to breakeven
- The level required to earn a specific amount of profit
- The effect of various cost and price assumptions

Uses of Cost-Volume Profit Analysis:

1. Show the impact of changes in selling price on required sales volume.
2. Reveal how anticipated cost changes affect profit levels.
3. Evaluate marketing strategies (advertising and promotion expenditures).
4. Assist in individual product pricing decisions.
5. Determine how much sales a new salesperson must generate to cover salary and costs.

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Breakeven Analysis Procedure

Breakeven point occurs when Total Revenue = Total Cost

Three Steps in Determining the Breakeven Point

Step 1: Classify Fixed and Variable Costs

Correct classification depends on understanding how the agribusiness operates.

Fixed Costs

Examples:

- Manager salary
- Depreciation of grain silos
- Insurance
- Rent
- Long-term loan interest

Variable Costs

Costs that **change directly with sales volume**.

Examples:

- Cost of goods sold
- Delivery fuel
- Commissions
- Animal feed

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Breakeven Analysis Procedure

Breakeven Analysis can be determined using the formula below:

(a) $$Breakeven = \frac{Fixed\ Cost}{\text{Price per unit} - \text{Variable Cost per Unit}}$$

Where:

$$(Price - Variable\ Cost) = Contribution\ Margin(CM)$$

(b) Contribution Margin (CM) represents:

The amount each unit contributes toward covering fixed costs and thereafter generating profit. $CM = P - AVC$

(c) Information Required to Compute Break-Even

To compute break-even, the agribusiness must know:

- Fixed Costs (FC)
- Selling Price per unit (P)
- Variable Cost per unit (AVC)

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Breakeven Analysis Procedure

EXAMPLE 1: Tomato Processing Small and Medium Enterprise (SME)

Given:

Fixed Cost = ZMW 50,000

Price per bottle = ZMW 25

Variable Cost per bottle = ZMW 15

$$\boldsymbol{Breakeven = \frac{Fixed\ Cost}{\text{Price per unit} - \text{Variable Cost per Unit}}}$$

Step 2: Compute Contribution Margin

$$\boldsymbol{CM = 25 - 15 = 10}$$

Step 3: Compute Break–Even Quantity

$$BEQ = \frac{50{,}000}{10} = 5{,}000\ bottles$$

Interpretation

Tomato processing firm needs to sell 5,000 bottles to recover its initial total cost.

Proof

TR = Price x Q

TR = 25 x 5,000 = K125,000

TC = FC + VC

TC = K50,000 + (15 x 5,000) = K125, 000

Profit = TR – TC

Profit = K125, 000 – K125, 000 = 0

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Breakeven Analysis Procedure

Profit Comparison at Different Output Levels

Option 1: Produce 4,000 Bottles

TR = 25 × 4,000 = 100,000

VC = 15 × 4,000 = 60,000

TC = 60,000 + 50,000 = 110,000

Profit = 100,000- 110,000 = –10,000 (Loss)

Decision: Not viable.

Option 2: Produce 5,000 Bottles

(Break–Even)

TR = 125,000

TC = 125,000

Profit = 0

Option 3: Produce 7,000 Bottles

TR = 175,000

VC = 105,000

TC = 155,000

Profit = 20,000

Decision: Economically viable.

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Sensitivity Analysis

Agribusiness decisions must consider price and cost changes

Scenario A: Increase in Variable Cost (Input Inflation)

Suppose tomato cost increases due to drought:

Variable Cost rises to 18.

New CM:

$$CM = 25 - 18 = 7$$

New BEQ:

$$BEQ = 50{,}000/7 = 7{,}143\ bottles$$

Risk increases: firm must sell more to survive.

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Target Sales Volume (Units) Required to Earn a Given Profit

Target profit analysis helps agribusiness managers determine the level of production and sales required to achieve a desired profit. It enables managers to set production and sales targets that will generate the desired profit level.

$$\text{Target } \textit{Output (Units)} = \frac{\textit{Fixed Costs} + \textit{Target F}}{\textit{Selling price} - \textit{Variable cos}}$$

EXAMPLE 1: Tomato Processing SME

Suppose the owner wishes to earn a **target profit of ZMW 30,000** during the month.

(a) Determine the **number of bottles that must be sold to achieve the target profit of ZMW 30,000**.

(b) Calculate the **total sales revenue** required to achieve the target profit.

(c) Briefly explain the importance of target profit analysis in agribusiness decision-making.

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Importance Break-even Analysis in Agribusiness

Break-even analysis is useful in:

- Poultry production
- Milling enterprises
- Dairy processing
- Agro-input dealerships
- Irrigation investments

Particularly important where:

- Input prices fluctuate
- Markets are seasonal
- Capital investment is high

Limitations of Break-even Analysis

1. Assumes constant price
2. Assumes constant variable cost per unit
3. Ignores risk and uncertainty
4. Based on short-run framework

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Class Practice (Tutorial) 1

KMU Maize Marketing Ltd. has the following information extracted from its books:

Fixed costs (per annum) = K10,000,000

Variable costs per metric tonne = K15,000

Selling price per metric tonne = K25,000

Normal sales volume (metric tonnes per month) = 100

Maximum capacity (bags per month) = 150

(a) Compute the break-even point

(b) What will be the impact of increasing labour costs by K5,000,000 per annum

(c) Compute additional sales needed to generate profits of K12,000,000 per annum. Is it feasible?

(d) Compute impact of reducing selling price by 10%

(e) Compute impact of increasing variable costs by 10% and reducing selling price by 5%

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(ii) Revenue and Cost Analysis

Revenue and cost decisions determine:

- How much to produce
- Whether production is profitable

Profit maximization occurs where the difference between total revenue and total cost is greatest. Thus, output decisions depend on **marginal analysis** under competitive market conditions.

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Revenue and Cost Concepts

(a) Marginal Revenue (MR)

$$MR = \frac{\Delta TR}{\Delta Q}$$

Additional revenue from selling one more unit of output.

In competitive agricultural markets:

$\boldsymbol{MR = Price}$ (Farmers are price takers)

(b) Marginal Cost (MC)

$$MC = \frac{\Delta TVC}{\Delta Q}$$

Additional cost of producing one more unit.

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Profit Maximization Rule

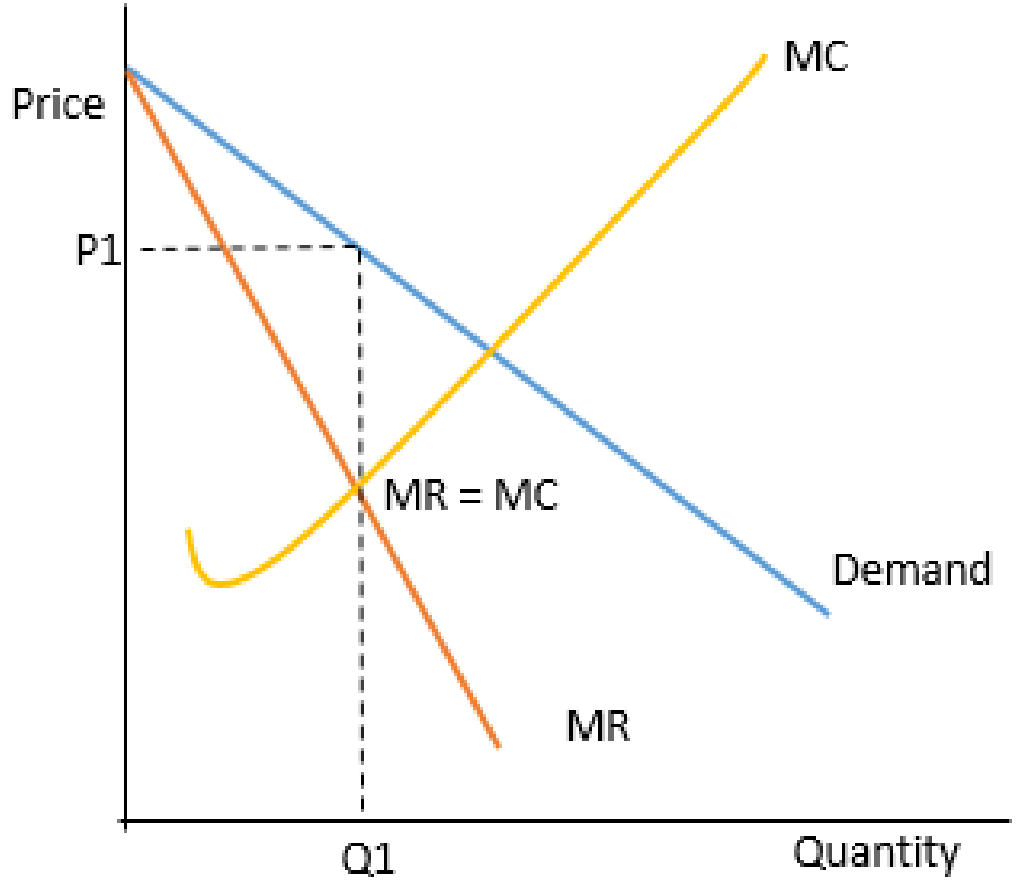
$MR = MC$

Decision rule:

- If MR > MC → Increase output
- If MR < MC → Reduce output
- If MR = MC → Optimal output level

Graph Interpretation:

- MC intersects MR from below
- Profit-maximizing output occurs at MR = MC
- ATC determines profit or loss



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EXAMPLE: Broiler Production Decision

A poultry farmer sells broiler chickens at a constant price of **K120 per bird**. The farmer wants to determine the profit-maximizing level of production. The farm incurs a fixed cost of **K40,000** per production cycle. The table below shows the estimated total variable costs (TVC) at different production levels.

Output (Birds)	Total Variable Cost (K)
500	45,000
1,000	85,000
1,500	132,500
2,000	190,000
2,500	262,500
3,000	355,000

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EXAMPLE: Broiler Production Decision

(a) Complete the table below by calculating:

- Total Revenue (TR)
- Total Cost (TC)
- Profit (π)
- Marginal Revenue (MR)
- Marginal Cost (MC)

Output (Birds)	TVC (K)	TR (K)	TC (K)	Profit (K)	MR (K)	MC (K)
500	45,000					
1,000	85,000					
1,500	132,500					
2,000	190,000					
2,500	262,500					
3,000	355,000					

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EXAMPLE: Broiler Production Decision

(a) Determine the output level that maximizes profit.

(b) Verify whether the profit-maximizing output satisfies the economic rule:

$$MR = MC$$

(c) Explain why producing beyond the profit-maximizing level may reduce profits.

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Solution: Broiler Production Decision

Step 1: We first collect the given data:

- Market price per bird = K120
- Fixed cost = K40,000

Step 2: (a) Calculate TR, TC, and Profit for **Option 1:** Produce **500 birds**

- $TR =$ $Price \times Quantity$ = 120 × 500 = K60,000
- TC = VC + FC = 45,000 + 40,000
- TC = K85,000
- **Profit** = TR - TC = K60,000 − K85,000
- **Profit** = - K25,000 (Loss)

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Solution: Broiler Production Decision

We Calculate Marginal Revenue (MR)

Formula:

$$MR = \frac{\Delta TR}{\Delta Q}$$

$$MR = \frac{\text{K60,000}}{500} = \text{K120}$$

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Solution: Broiler Production Decision

We Calculate Marginal Cost (MC)

Formula:

$$MC = \frac{\Delta TVC}{\Delta Q}$$

$MC =$ K85,000-K45,000 / 1000 - 500 = K80

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Solution: Broiler Production Decision

Output	TVC (K)	TR (K)	TC (K)	Profit (K)	MR (K)	MC (K)
500	45,000	60,000	85,000	-25,000	120	-
1,000	85,000	120,000	125,000	-5,000	120	80
1,500	132,500	180,000	172,500	7,500	120	95
2,000	190,000	240,000	230,000	10,000	120	115
2,500	262,500	300,000	302,500	-2,500	120	145
3,000	355,000	360,000	395,000	-35,000	120	185

Identify the Profit-Maximizing Output

- Profit Maximization occurs where MR=MC
- Thus, profit is maximized at approximately **2,000** birds, where MC is closest to MR (MR≈MC).
- Beyond this point, the cost of producing an extra bird exceeds the revenue earned from selling it.
- Producing beyond 2,000 birds reduces profit.

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Solution: Broiler Production Decision

Final Answer

Profit-Maximizing Output =2,000 birds

Maximum Profit =K10,000

Profit is maximized where MR≈MC

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To be Continued

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